Experiment 02

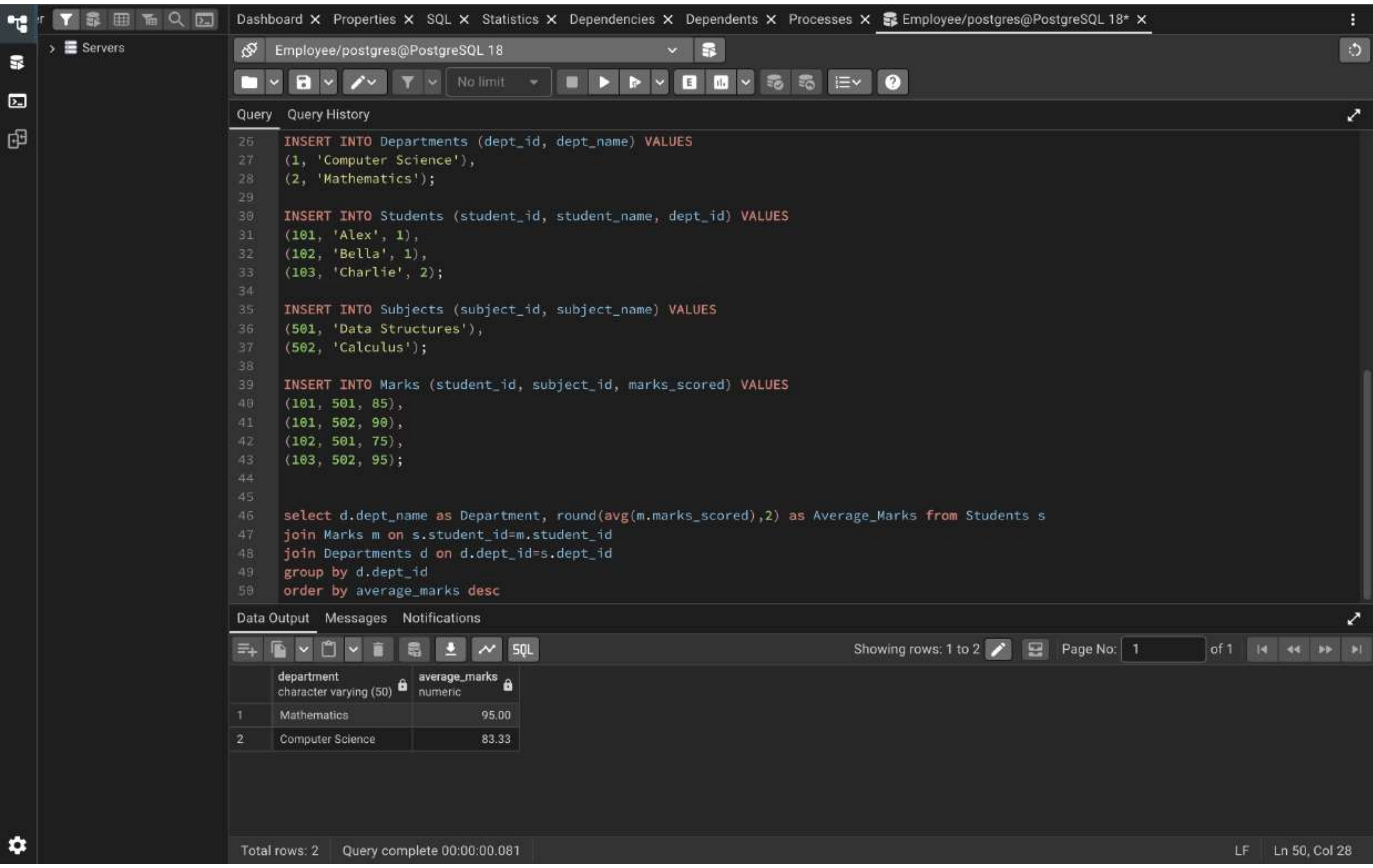
Level: Easy

- **(A)**

  A university administration requires an analytical performance report for each academic unit. Write a PostgreSQL solution to calculate the average marks scored by students grouped by their respective departments. The solution must join the student profile, course subjects, and grade marks tables, returning a breakdown sorted by the highest average marks down to the lowest.

Output

Department	Average Marks
Mathematics	95.00
Computer Science	83.33


Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
Processes
Employee/postgres@PostgreSQL 18*
Servers
Employee/postgres@PostgreSQL 18
No limit
Query
Query History
INSERT INTO Departments (dept_id, dept_name) VALUES
(1, 'Computer Science'),
(2, 'Mathematics');
INSERT INTO Students (student_id, student_name, dept_id) VALUES
(101, 'Alex', 1),
(102, 'Bella', 1),
(103, 'Charlie', 2);
INSERT INTO Subjects (subject_id, subject_name) VALUES
(501, 'Data Structures'),
(502, 'Calculus');
INSERT INTO Marks (student_id, subject_id, marks_scored) VALUES
(101, 501, 85),
(101, 502, 90),
(102, 501, 75),
(103, 502, 95);
select d.dept_name as Department, round(avg(m.marks_scored),2) as Average_Marks from Students s
join Marks m on s.student_id=m.student_id
join Departments d on d.dept_id=s.dept_id
group by d.dept_id
order by average_marks desc
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 2
Page No: 1
of 1
department character varying (50)
average_marks numeric
1 Mathematics 95.00
2 Computer Science 83.33
Total rows: 2
Query complete 00:00:00.081
LF
Ln 50, Col 28

Experiment 02

Level: Medium

Two legacy HR systems (A and B) have separate records of employee salaries. These records may overlap. Management wants to **merge these datasets** and identify **each unique employee** (by EmpID) along with their **lowest recorded salary** across both systems.

Objective

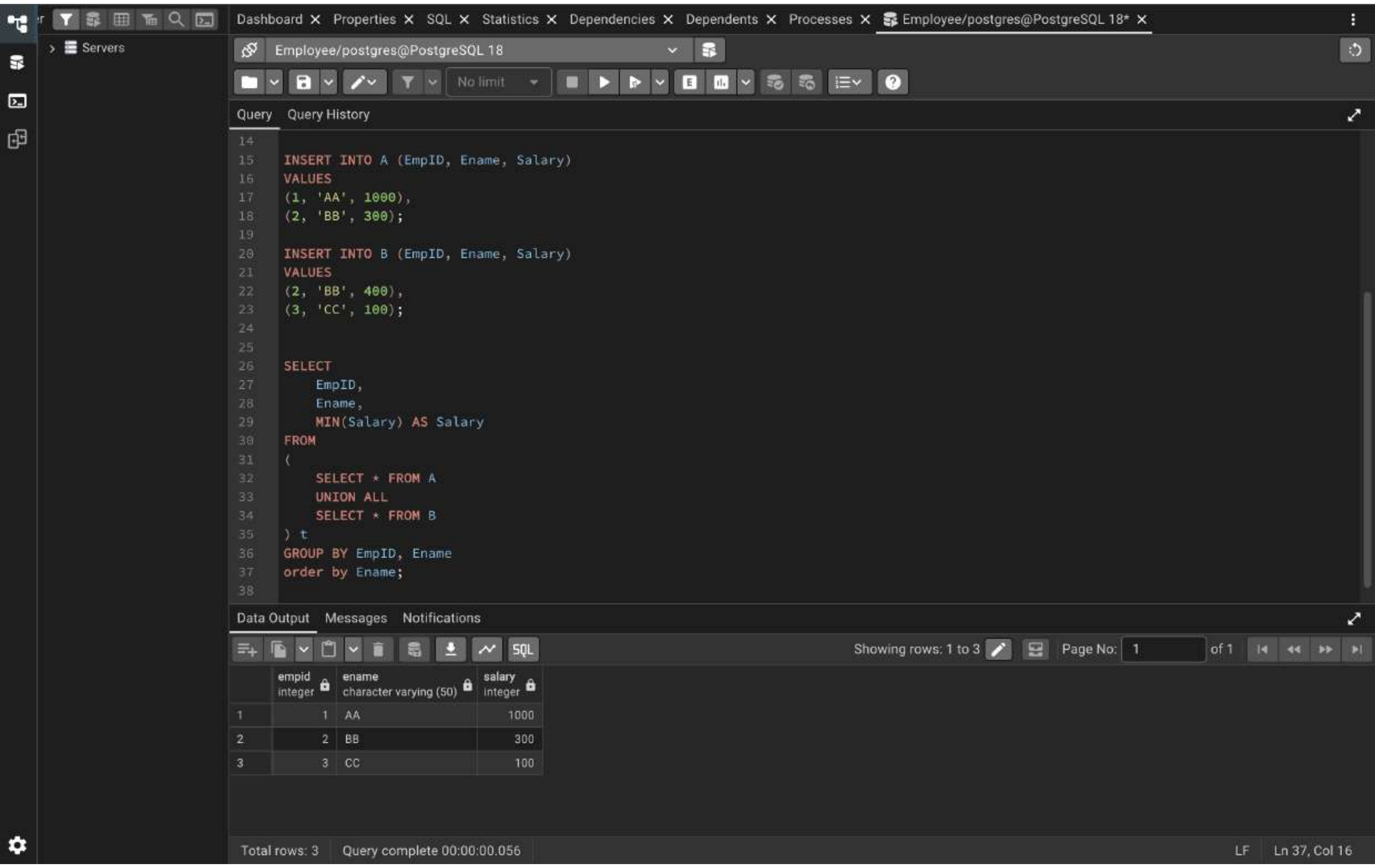
1. Combine two tables A and B.
2. Return each EmpID with their **lowest salary,** and the corresponding **Ename.**

Table A

EmpID	Ename	Salary
1	AA	1000
2	BB	300

Table B

EmpID	Ename	Salary
2	BB	400
3	CC	100



Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
Processes
Employee/postgres@PostgreSQL 18*
Servers
Employee/postgres@PostgreSQL 18
No limit
Query
Query History
INSERT INTO A (EmpID, Ename, Salary)
VALUES
(1, 'AA', 1000),
(2, 'BB', 300);
INSERT INTO B (EmpID, Ename, Salary)
VALUES
(2, 'BB', 400),
(3, 'CC', 100);
SELECT
EmpID,
Ename,
MIN(Salary) AS Salary
FROM
(
SELECT * FROM A
UNION ALL
SELECT * FROM B
) t
GROUP BY EmpID, Ename
order by Ename;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 3
Page No: 1
of 1
empid integer
ename character varying (50)
salary integer
1 1 AA 1000
2 2 BB 300
3 3 CC 100
Total rows: 3
Query complete 00:00:00.056
LF
Ln 37, Col 16

Experiment 02

Level: Hard

A multinational organization maintains separate records for employee information and department details. The management wants to identify the employees who receive the highest salary within their respective departments for an annual performance report.
If more than one employee earns the same highest salary in a department, all such employees should be included in the report. The final output should display the **Department Name**, **Employee Name**, and **Salary**, arranged in ascending order of department name.

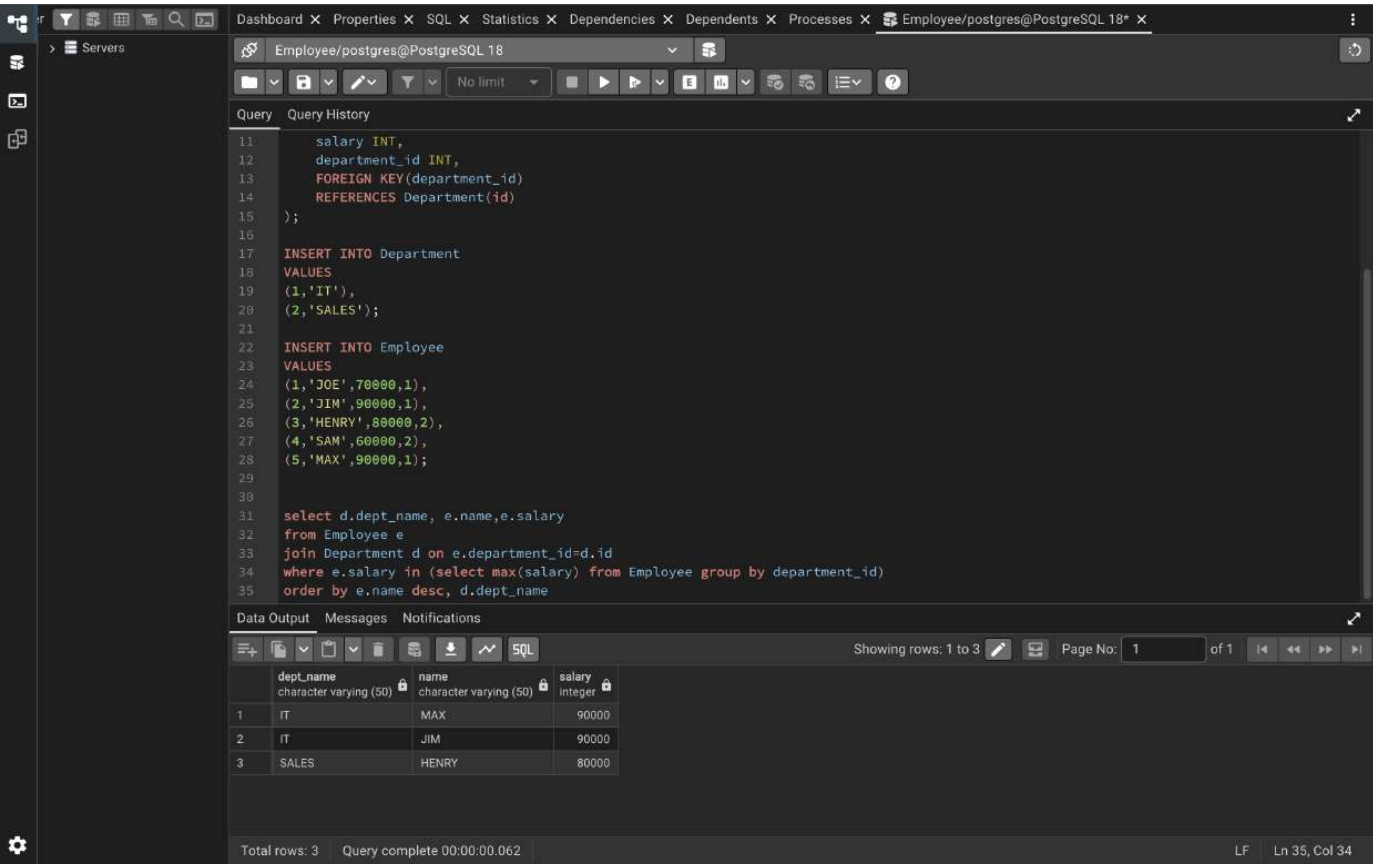
Note: The employee and department information are maintained in separate normalized tables.

Input Table: **Employee**

ID	NAME	SALARY	DEPT_ID
1	JOE	70000	1
2	JIM	90000	1
3	HENRY	80000	2
4	SAM	60000	2
4	MAX	90000	1

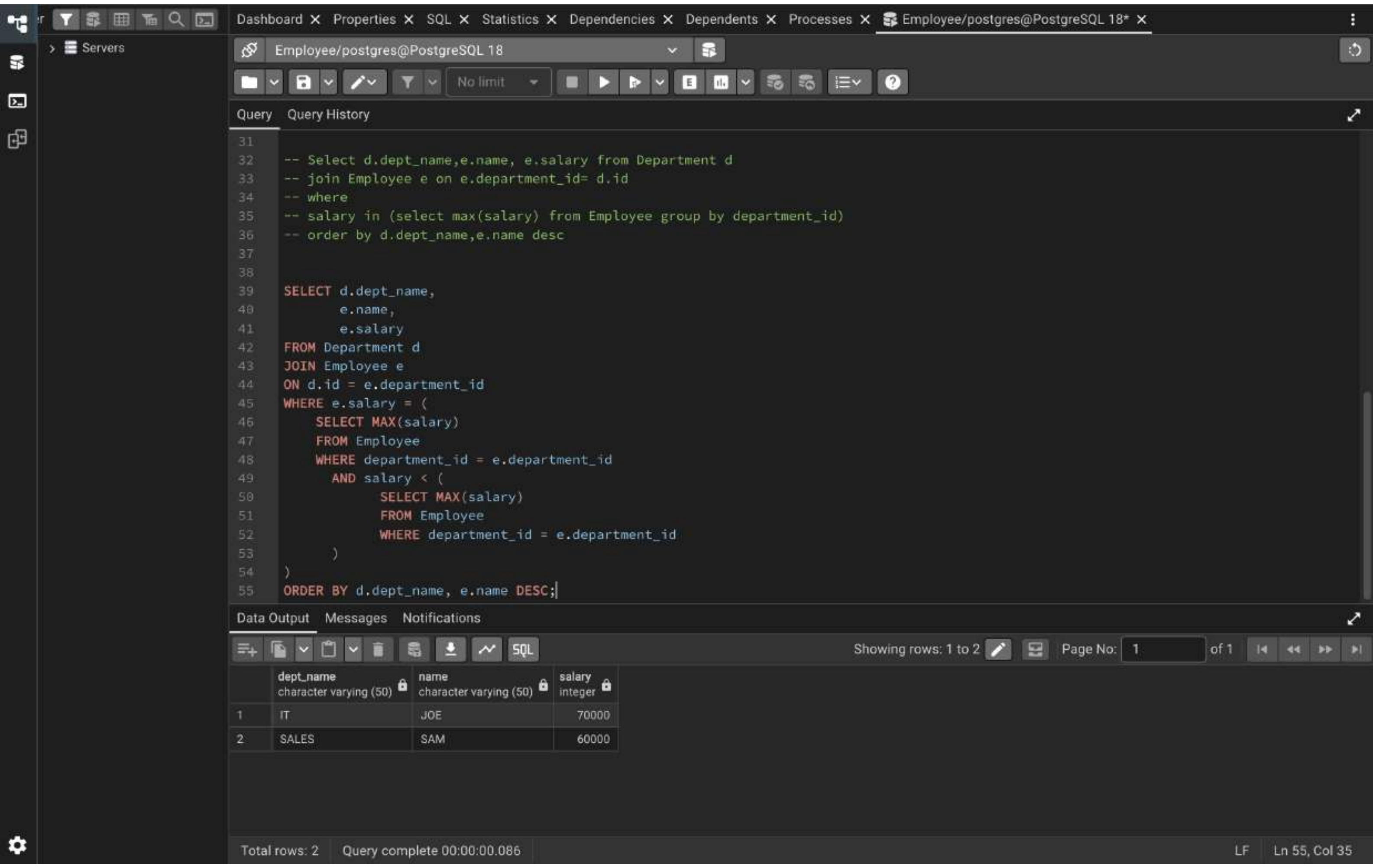
Department

ID	DEPT_NAME
1	IT
2	SALES


Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
Processes
Employee/postgres@PostgreSQL 18*
Servers
Employee/postgres@PostgreSQL 18
No limit
Query
Query History
salary INT,
department_id INT,
FOREIGN KEY(department_id)
REFERENCES Department(id)
);
INSERT INTO Department
VALUES
(1,'IT'),
(2,'SALES');
INSERT INTO Employee
VALUES
(1,'JOE',70000,1),
(2,'JIM',90000,1),
(3,'HENRY',80000,2),
(4,'SAM',60000,2),
(5,'MAX',90000,1);
select d.dept_name, e.name,e.salary
from Employee e
join Department d on e.department_id=d.id
where e.salary in (select max(salary) from Employee group by department_id)
order by e.name desc, d.dept_name
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 3
Page No: 1
of 1
dept_name character varying (50)
name character varying (50)
salary integer
1 IT MAX 90000
2 IT JIM 90000
3 SALES HENRY 80000
Total rows: 3
Query complete 00:00:00.062
LF
Ln 35, Col 34

Experiment 02

Part B: FIND THE 2ND HIGHEST EARNING EMPLOYEE from the same schema.


Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
Processes
Employee/postgres@PostgreSQL 18*
Servers
Employee/postgres@PostgreSQL 18
No limit
Query
Query History
-- Select d.dept_name,e.name, e.salary from Department d
-- join Employee e on e.department_id= d.id
-- where
-- salary in (select max(salary) from Employee group by department_id)
-- order by d.dept_name,e.name desc
SELECT d.dept_name,
e.name,
e.salary
FROM Department d
JOIN Employee e
ON d.id = e.department_id
WHERE e.salary = (
SELECT MAX(salary)
FROM Employee
WHERE department_id = e.department_id
AND salary < (
SELECT MAX(salary)
FROM Employee
WHERE department_id = e.department_id
)
)
ORDER BY d.dept_name, e.name DESC;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 2
Page No: 1
of 1
dept_name character varying (50)
name character varying (50)
salary integer
1 IT JOE 70000
2 SALES SAM 60000
Total rows: 2
Query complete 00:00:00.086
LF
Ln 55, Col 35